Galata Sea and Migration Museum: an immersive and interactive visitor experience

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Abstract: This project presentation focuses on new visitor experiences developed by the Galata Sea and Migration Museum (Mu.MA) in Genoa (Italy) in partnership with ETT S.p.A. ICT company.

This collaboration has already produced several permanent exhibitions, where new technologies better "explain the message" that the museum wants to impart. The first was the "Nazario Sauro" pre-show, an exhibition that reveals life in a submarine. Then came the Memories and Migration (MEM) section, which tells the history of emigration from, and immigration to, Italy by presenting real migrant stories. On a fully recreated 17th century Genoese galley, visitors enjoy an interactive experience with the "crew". The new Christopher Columbus section has portraits and video-audio renditions of his documents in various languages.

These exhibitions integrate original content with innovative solutions such as high-resolution monitors, interactive touch-screen interfaces, 3D reconstructions, proximity sensors, gaming simulations, augmented reality applications and interactive projections with on-screen actors - ensuring complete visitor immersion, and promoting and renewing interest and involvement. The museum aims to revitalise communication with visitors, transforming them from passive to active through an "immersive" experience. This approach is based on the concepts of "knowing through doing" and "recognise by touching", creating synergy with gallery artwork and environments, so that each age, academic and social group (from school children through to students, adults, parents and pensioners) will dynamically relate to the immersive visitor experience. Galata Museum's renewal action was carried out creating meticulous cooperation between museum curators (original content and context) and ETT (experts in the fields of ICT interactivity and museology). This collaboration works when the two main objectives are satisfied: 1) enhancement of inherent value of content and context; 2) rapid return on investment through substantial visitor growth - especially true now that central government funding is progressively less.

Keywords: migration museum, interactive museum exhibition, gaming simulation, touch screen interface, 3D reconstruction, augmented reality, Galata Sea and Migration Museum, ETT S.p.a.

Galata Museum – ETT cooperation: a new museum approach

This project presentation focuses on a new concept of visitor experience, developed by the **Galata Sea and Migration Museum (Mu.MA) in Genoa (Italy),** in partnership with **ETT S.p.A.**, an ICT company. The partnership - started in 2009 and still running - has helped Mu.MA to revitalise its exhibition configuration through the use of innovative multimedia devices, aimed at delivering the museum's message in a more efficient and inspiring way. Thanks to the many technological ICT advances developed in the last few years, the museum has been able to apply new and modern approaches to enhancing the visitor experience, while conserving historical context and content.

Galata Sea and Migration Museum

Opened in 2003, the Galata Sea Museum is the Mediterranean's largest and most innovative maritime museum. It is notable in Italian maritime museum design for the quality and innovation of its creations. Today, with an assorted audience made up of people of differing ages and education, and often living far from the sea and its culture, Galata makes an impact though the ability to immerse the visitor in meticulous high-quality scientific reconstructions that give the best idea of an era, type of ship, and on-board life.



Galata Sea Museum building in Genoa

Travel is one of the main themes covered during a visit to the Galata Sea Museum: voyages which, while taking advantage of technical advances in transport methods, have always tempted man to challenge seas and oceans. The journey through the Museum, in chronological order, starts on the ground floor in the age of oar-powered ships, continues on the first and second floors on the old sailing routes of revolutionary geographical exploration, and ends on the third floor dedicated to the voyage to the Americas.

The Galata Sea Museum has chosen to devote a significant portion of its space to the permanent section entitled MEM (Memory and Migration). This is the first permanent museum section in Italy dedicated to “migration”. The task of MEM is to “remember” national migration history and promote, through an international network of regional museums, a “local” memory of migration. Local museums are becoming increasingly important in this respect, and are helping communities to reinforce and even rediscover their identities. Of particular importance is the social cohesion effect, especially in areas containing minority groups among the population. There are now many Jewish and anti-racist museums across Europe, while other museums have dedicated sections to “emigration” and “immigration”. Important among these are Genoa, Bremerhaven, Paris, Cherbourg, Antwerp and Belfast.

The immersive approach and the use of technology

In recent years, MUMA has established a definite preference for multimedia and interactive tools, which present content to the public in an active way (as opposed to the traditional passive way) through an “**immersive**” experience. This approach is based on the concepts of “knowing through doing” and “recognise by touching”, and on creating synergy with artwork and gallery environments. In this way, each age, academic and social group (from school children through to students, adults, parents and pensioners) will dynamically relate to the immersive visitor experience. The visitor shouldn’t be impartial or neutral, but must be torn from his private world and transported to another era. Interactive tools enable content to be offered to visitors having different levels of understanding; both children and adults should come away with something positive in mind.

To offer an ever new and engaging experience to its visitors, the Galata Sea Museum has elected to progressively renew its permanent installations by using the best technological tools available on the market. The technologies used must be intuitive and easy to use, allowing operation by visitors of all ages and with different levels of familiarity with technological tools. They must also be sufficiently robust, both to resist visitor use over time, and to enable new content to be added for future visits.



Emigration multimedia immersive exhibition

Prominent tools in current use are HD and 3D video content (both with glasses and auto-stereoscopic), integrated and interactive new-media software content, touch and multi-touch screen technology, augmented reality, gaming simulation, hologram technology, RFID (Radio Frequency IDentification), voice or movement activated features, and QR (Quick Response) code connections.

A new 4D installation (including synchronised physical effects) is being developed and will be presented in the near future. Visitors will find themselves in the midst of a **sea storm** and will be able to appreciate how it feels to be at the mercy of the elements.

MUMA multimedia installations

This collaboration has already produced several permanent exhibitions, where new technologies better "explain the message" that the museum wants to impart.

The first was the **"Nazario Sauro" pre-show**, inaugurated in May 2010, an exhibition that combines a multimedia area, where life on a submarine is shown in all its shades, before a visit on board the real submarine berthed in the dock just outside the museum. The multimedia area is not a "war game" but comprises the physical reconstruction of a submarine environment (settings, fittings and sound) and a learning trail where visitors discover the history of submarines, listen to stories told by the crew, explore the interior of the submarine using 360 degree 3D imaging, experiment with the Archimedes principle[1], use the hydrophone and periscope, and drive the submarine on an interactive simulator. At the end of the "course", a personalised submariner's diploma is issued.

[1] The various buoyancy tanks in a submarine, which determine how it floats, dives and surfaces, derive from the Archimedes Principle. Archimedes of Syracuse (c. 287 BC – c. 212 BC) discovered that, "any floating object displaces its own weight of fluid" and "any object, wholly or partially immersed in a fluid, is buoyed up by a force equal to the weight of the fluid displaced by the object". Using this principle, we can:

- measure the volume of an object by measuring the volume of the liquid it displaces after submerging;
- calculate the buoyancy of an object immersed into a liquid.



Submarine "Nazario Sauro" pre-show

The **Memory and Migration** (MEM) section, opened in November 2011, tells the history of emigration from, and immigration to, Italy by presenting real stories using over forty multimedia installations. The exhibition begins with ***Emigration*** (the past), where visitors experience a voyage departing from Genoa on transatlantic liners. Immigrants sailing to Ellis Island (New York), Brazil, or Argentina give first-hand testimonies, writings and photos. Each visitor receives a personal passport at the beginning of the voyage, which is then used at each interactive stage of the journey. The results of tests, plus the answers given at an immigration interview, will affect the outcome. All this gives visitors an opportunity to reflect on the experiences of immigrants. The area dedicated to ***Immigration*** (the present) shows the stories and the travels of migrants to Italy. This movement began in the 1990s and is still far from over. It has cost the lives of tens of thousands of people in the Straits of Sicily alone (about 1,500-2,000 deaths were calculated in just one year, 2011, the year of the Arab Spring). The installation dedicated to the 2011 landings on the Sicilian island of Lampadusa also includes two "Lampedusa boats", saved from destruction. Lampedusa was recently in the news again, when more than 500 men, women and children, mostly from Eritrea and Somalia, were on a boat that capsized close to the island. Only 155 survived. Other multimedia installations offer some examples of multicultural society in workplaces and at school, as well as cooking recipes from the five continents.



MEM – Memory and migration section

The **Desks of Knowledge** are new multimedia stations that enable visitors to discover and enrich their history of the Crusades and of the ancient order of the Knights Hospitallers of St. John of Jerusalem, who founded the "Commenda - Hospital in Prè", an architectural monument dating back to the 12th century A.D. The multimedia tools reproduce an exciting journey through the Middle Ages, retracing the routes and everyday lives of pilgrims, testified to by the travel stories of both famous and lesser-known characters, such as Saladino, Caffaro of Caschifellone III, Beniamino of Tudela, Benedetto Zaccaria and Matthew Paris.

Climb Aboard the Galley opened in June 2012. Visitors climb aboard a full-sized reconstruction of a seventeenth century Genoese galley and enjoy an interactive experience with the "crew", choosing to be a slave, a prisoner on forced labour, or a "buonavoglia", a volunteer probably paying off a debt. They participate in discussions between the senator, the captain and the shipwright, and discover aspects of life on board in the galley's "games room". This journey through time is completed by admiring rare antiquities and works.

The new **Christopher Columbus** section contains portraits, audio-video renditions of his documents, including the precious Code of Privileges, in various languages: mediaeval Latin, old Italian or 16th century Spanish - Columbus's language of choice. Visitors use a touch-screen to explore a painting by Cristoforo Grassi, *View of the City of Genoa in 1481,* one of the oldest and most precise indications of the shape of the port and city of Genoa in the fifteenth century. Additional content may be discovered using QR codes in three languages.



Ancient port of Genoa touch screen interface

Public/private teamwork benefits everyone

The Galata Museum revitalisation action is successful due to meticulous cooperation between museum curators (original content and context) and ETT (experts in the fields of ICT interactivity and museology). Complete solutions typically comprise four stages:

- initial concept and design;
- content architecture;
- museology and installation (multimedia showcases, stations, videos etc.);
- management and maintenance.

Cost efficient services cover the entire process creating:

1) added value of content and context;

2) rapid return on investment through substantial visitor growth.

Central government funding is progressively less and additional visitor revenue is becoming increasingly important. When astute investment enhances the inherent value of museum content, the cost is amply recovered through additional visitors or users.

Since the museum adopted the new approach to the “Immersive and Interactive Visitor Experience”, in 2009, it has progressively invested around €400,000 and the annual number of visitors has grown significantly, from 80,000 in 2006, to 130,000 in 2009, 230,000 in 2011, and over 250,000 in 2012.

Best practices

MuMa museum has been classified as a multimedia excellence among Italian museums by Skål International (Rome), the largest organization of professionals and tourism leaders around the world.

In September 2013, The Galata Sea and Migration Museum participated at the **MeLa Project** (“***European Museums in an Age of Migrations*”** project funded by the European Commission under the Socio-economic Sciences and Humanities Program - FP7th) mid-term seminar “Let the Museum Speak: European Museums in an Age of Migrations” with a paper and presentation.

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References

Galata Museum website:http://www.galatamuseodelmare.it/

Nazario Sauro submarine website: http://www.sommergibilenazariosauro.it/

Case histories from ETT. website:

- Climb aboard the Galley: http://www.ettsolutions.com/ProjectDetails.aspx?id=166
- Remembrance and Migration: http://www.ettsolutions.com/ProjectDetails.aspx?id=147
- The Desks of Knowledge: http://www.ettsolutions.com/ProjectDetails.aspx?id=164
- Columbus: http://www.ettsolutions.com/ProjectDetails.aspx?id=206

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